Zixuan Chen

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Education

Multimedia Laboratory (MMLab), The Chinese University of Hong Kong Aug. 2025 – Present
Ph.D. in Information Engineering Supervisor: Prof. Tianfan Xue

Zhejiang University Sept. 2022 – Mar. 2025
M.Eng. in Aeronautical and Astronautical Science and Technology Supervisor: Prof. Zheming Lu

Zhejiang University Sept. 2018 – Jun. 2022
B.E. in Flight Vehicle Design and Engineering GPA: 3.86/4.0 Rank: Top 10%

Research Interests

Generative Models, Computational Photography

Research Experience

Ph.D. Student **MMLab, The Chinese University of Hong Kong** Aug. 2025 – Present

- **Topic:** Drift-resistant multi-turn image editing by steering the tail of pretrained editors' denoising trajectories.
- **Topic:** Generative image stitching under large parallax using bi-level 3D projected RoPE.
- **Topic:** Ultra-high dynamic range (HDR) imaging formulated as a guided inpainting problem.
- Worked with Prof. Tianfan Xue.

Master's Student **Zhejiang University** Sept. 2022 – Mar. 2025

- **Topic:** Lightweight image dehazing network based on the re-parameterizable detail-enhanced convolution (DEConv).
- **Topic:** Training-free and model-agnostic domain adpatation for image dehazing using statistical prompt.
- Worked with Prof. ZheMing Lu and Prof. Zewei He.

Publications

- Ruikang Li, **Zixuan Chen**, Jiarui Wu, Mingde Yao, Yiyuan Zhang, Yujin Wang, Hanming Deng, Lewei Lu, Tianfan Xue. "TailSteer: Trajectory-Tail Steering for Drift-Resistant Multi-turn Image Editing." *Under Review*, 2026.
- **Zixuan Chen**, Lingen Li, Junhao Zhuang, Ruikang Li, Tianfan Xue. "AnyStitch: Generative Large-Parallax Image Stitching." **ACM Transactions on Graphics (TOG)**, SIGGRAPH Asia, 2026. [paper]
- Yichen Bian, **Zixuan Chen**, Yujin Wang, Xin Cai, Shi Guo, Junhao Zhuang, Leilei Gu, Tianfan Xue. "SwiftFusion: Towards Motion-Robust and Efficient 4K HDR Fusion." **ACM Transactions on Graphics (TOG)**, SIGGRAPH Asia, 2026. [code]
- Xingyu Liu, Zewei He, Yu Chen, Chunyu Zhu, **Zixuan Chen**, Xing Luo, Zhe-Ming Lu. "Unified Removal of Raindrops and Reflections: A New Benchmark and a Novel Pipeline." European Conference on Computer Vision (ECCV), 2026. [paper, code]
- Yu Chen, Zewei He, Xingyu Liu, **Zixuan Chen**, Zhe-Ming Lu. "GFRRN: Explore the Gaps in Single Image Reflection Removal." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [paper, code]
- **Zixuan Chen**, Yujin Wang, Xin Cai, Zhiyuan You, Zhe-Ming Lu, Fan Zhang, Shi Guo, Tianfan Xue. "UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **Highlight, Best Demo Honorable Mention**, 2025. [paper, code]
- **Zixuan Chen**, Zewei He, Ziqian Lu, Xuecheng Sun, Zhe-Ming Lu. "Prompt-based Test-time Real Image Dehazing: A Novel Pipeline." European Conference on Computer Vision (ECCV), 2024. [paper, code]
- **Zixuan Chen**, Zewei He, Zhe-Ming Lu. "DEA-Net: Single Image Dehazing Based on Detail-Enhanced Convolution and Content-Guided Attention." IEEE Transactions on Image Processing (TIP), **1K+ citations**, 2024. [paper, code]

Honors and Awards

CUHK Vice-Chancellor's PhD Scholarship 2025
Outstanding Graduate of Zhejiang Province (Top 5%) 2022
National Scholarship (Top 3%) 2021
Zhejiang University Scholarship – First Prize (Top 5%) 2021

Services

Conference Reviewer: CVPR, ICCV, ECCV, NeurIPS, AAAI (SPC), ACMMM, ACMMM Asia
Journal Reviewer: IEEE TPAMI, IEEE TVCG, IEEE TIP, IEEE TMM